Zehranaz Canfes

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Education

Master of Science | Computer Science | October 2022 - March 2026
***Technical University of Munich**, Munich, Germany*

Bachelor of Science (Double Major) | Computer Engineering | September 2018 - June 2022
***Bogazici University**, Istanbul, Turkey*
GPA: 3.45/4.00

Bachelor of Science (Double Major) | Mathematics | September 2017 - June 2022
***Bogazici University**, Istanbul, Turkey*
GPA: 3.45/4.00

Publications

TwoSquared: 4D Generation from 2D Image Pairs, **International Conference on 3D Vision (3DV) 2026,**
Lu Sang*, **Zehranaz Canfes***, Dongliang Cao, Riccardo Marin, Florian Bernard, Daniel Cremers
Access paper here.

4Deform: Neural Surface Deformation for Robust Shape Interpolation, **Conference on Computer Vision and Pattern Recognition (CVPR) 2025**,
Lu Sang, **Zehranaz Canfes**, Dongliang Cao, Riccardo Marin, Florian Bernard, Daniel Cremers
Access paper here.

Implicit Neural Surface Deformation with Explicit Velocity Fields, **International Conference on Learning Representations (ICLR) 2025**,
Lu Sang, **Zehranaz Canfes**, Dongliang Cao, Florian Bernard, Daniel Cremers
Access paper here.

Text and Image Guided 3D Avatar Generation and Manipulation, *Spotlight* on **IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2023**,
Zehranaz Canfes, M. Furkan Atasoy, Alara Dirik, Pinar Yanardag
Access paper here.

Experience

Computer Vision Student Researcher | May 2025 - Present
Computer Vision Group, Technical University of Munich, Munich, Germany | cvg.cit.tum.de

- Conducting **research** on uncovering the feature space of large 3D generative models to analyze emergent **correspondence patterns** and **semantics**.

Generative AI Researcher | Internship | April 2024 - October 2024
The BMW Group, Munich, Germany | bmwgroup.jobs

- **Trained** and **tested** state-of-the-art 3D generative models and their latent space using different surface representation methods such as **B-reps, NURBS, point clouds** by using **Python**, **PyTorch**, **PythonOCC**, **occwl**, and **geomdl**. The results are used for further research in BMW.

Undergraduate Researcher | October 2021 - June 2022
Creative AI Technologies Research Lab, Istanbul, Turkey | catlab-team.github.io

- **Published a paper** on 3D avatar editing guided by text or images by manipulating the latent space of a 3D generative network at the **WACV 2023 conference**. The model is implemented using **Python**, **Tensorflow**, and **PyTorch,** and achieves **34% higher scores** than previous approaches.

Artificial Intelligence Researcher | Internship **|** July 2021 - September 2021
Università di Bologna, Bologna, Italy | corsi.unibo.it

- **Proposed a neural network architecture** (autoencoder model) to detect anomalies in a semi-supervised way by using **Python** and **Tensorflow**. The proposed architecture **increased the F2-score by 30%.**

Class Projects

Guided Research and Interdisciplinary Project (IDP)
- Contributed to **three research projects** and **academic papers** on deformable **3D/4D** generation with diffusion models, resulting in multiple academic publications.

Computer Vision Practical Course
- Participated in the practical course: Shape Reconstruction and Matching in Computer Vision at the TUM Computer Vision Group. **Improved** an existing approach to work on **multi-view 3D reconstruction** of objects with non-trivial backgrounds by using **Python**, **Pytorch**, and **Pytorch3D.**

Machine Learning for 3D Geometry
- Adapted the codebase of the paper 3D-LMNet, which originally used **TensorFlow 1.3**, to **PyTorch**. The project involved ensuring that the code maintained its original functionality and performance for the task of **single-view reconstruction** of 3D point clouds.The code can be found here.

Certificates and Awards

Scholarship | DAAD-TEV
DAAD-TEV-Master's Degree Scholarship

IELTS
8.0/9.0

German Language Certificate | Sprachdiplom Kultusministerkonferenz
Level II, C1